

Annex B

Cover Page for College to submit with Prelim Examination Paper (NJC – 2009 Prelim H2) :

Qn/No	Topic Set	Paper Number	Answers	Marks
1	Inequalities	9740/01	$x \leq -5$ or $-2 < x \leq 3$; $-\sqrt{3} \leq x \leq \sqrt{3}, x \neq 0$	5
2	Functions	9740/01	(i) $f^{-1} : x \mapsto \sqrt{4-x}, D_{f^{-1}} = (-\infty, -5]$ (ii) $g(x) = 11 - 2x ; a \geq -5$	6
3	Vectors	9740/01	(i) $3\sqrt{3}$ (ii) $\overrightarrow{AN} = \begin{pmatrix} 2-2\lambda \\ 3 \\ 1+2\lambda \end{pmatrix} ; \lambda = \frac{1}{4}$	6
4	Complex Numbers (Loci)	9740/01	(ii) $a = \sqrt{2}$	6
5	Differential Equations	9740/01	$y = x - Axe^{-x}$	6
6	Method of Difference	9740/01	(i) $\frac{1}{4} - \frac{1}{2N(N+1)}$ (ii) $\frac{1}{180}$	7
7	Differentiation	9740/01	(a) $\frac{dx}{dt} \approx 0.196 \text{ cm/s}$ (b) $(-\frac{2p}{\sqrt{5}}, \frac{p}{\sqrt{5}})$ and $(\frac{2p}{\sqrt{5}}, -\frac{p}{\sqrt{5}})$	8
8	Integration (Techniques)	9740/01	(a) $A = \frac{1}{2}$ and $B = -2$; $\ln \sqrt{x^2 + 4x + 6} - \sqrt{2} \tan^{-1} \left(\frac{x+2}{\sqrt{2}} \right) + c$ b(ii) $-\sin^{-1} \left(\frac{1}{x} \right) + C$ or $\cos^{-1} \left(\frac{1}{x} \right) + C$; $\left(\frac{-1}{\sqrt{x^2 - 1}} \right) \ln x - \sin^{-1} \left(\frac{1}{x} \right) + C$ OR $\left(\frac{-1}{\sqrt{x^2 - 1}} \right) \ln x + \cos^{-1} \left(\frac{1}{x} \right) + C$	11

9	Complex Numbers (Algebra)	9740/01	(a) $z = 2e^{i\frac{5\pi}{6}}$; $a = 4\ln 2$, $b = -\frac{2\pi}{3}$ (b) $w = -2, 1+i\sqrt{3}, 1-i\sqrt{3}$; $w = -1, -2, 1+i\sqrt{3}, 1-i\sqrt{3}$; $z = i, 2i, \sqrt{3}-i, -\sqrt{3}-i$	11
10	Vectors (lines/planes)	9740/01	(i) $\vec{r} \cdot \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = 1$ (ii) $\overrightarrow{ON} = \begin{pmatrix} 3 \\ -2 \\ -2 \end{pmatrix}$ (iii) $\vec{r} = \begin{pmatrix} 3 \\ -2 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ (v) $D = 20$	13
11	AP,GP	9740/01	(ii) $0 < x < 0.354$ (iii) $a \approx 191$	9
12	Curve Sketching	9740/01	(ii) Vertical asymptote: $x = 5$ Oblique asymptote: $y = kx + 5k$ (iii) $(1, 2k)$ and $(9, 18k)$	12
Pure Maths				
1	Binomial Series	9740/02	$k = 9$; $-\frac{1}{3} < x < \frac{1}{3}$	4
2	System of Linear Equations	9740/02	1:3:1	4
3	Maclaurin's Series	9740/02	$x + x^2 - \frac{1}{6}x^3$; $1 + x - \frac{3}{2}x^2$	7
4	Mathematical Induction; Recurrence	9740/02	a(i) $u_2 = 18$; $u_3 = 96$; $u_4 = 600$ (ii) Conjecture: $u_n = (n+1)^2 n!$ b (i) $a = -1.373$, $b = 1.924$	13
5	Integration (Area/volume)	9740/02	a(iii) 5 (b) $(1, 3)$ and $(2, 0)$; 4.05	12
Statistics				
6	Sampling method	9740/02	(i) Quota sampling	4
7	Hypothesis Testing	9740/02	21.9 ; $\alpha > 4.25$	6
8	P & C	9740/02	(a) 36 b(i) 720 (ii) 3456	7
9	Binomial/ Poisson/Sampling	9740/02	(i) 0.323 (ii) 0.500 ; 0.210	8
10	Prob	9740/02	(i) $\frac{1}{31}$ (ii) $\frac{11}{24}$ (iii) least $n = 17$	10
11	Regression	9740/02	$x = 10, y = 42$ (i) 7.4 (iv) $c = 402$	10
12	Normal	9740/02	$\mu = 2.5$ (ii) 0.0755 (iii) 0.00393 ; $a = 2.72$ kg	12